

Haripriya P

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github.com/haripriyap07 | [personal-portfolio](#)

Summary

Robotics & Automation Engineering student (CGPA 8.9/10) specializing in robust backend infrastructure with Java and Spring Boot. Combines systems-level engineering principles with hands-on experience building production-style web systems, caching strategies, and containerized deployments. Seeking Backend Engineering / SWE internship roles.

Technical Skills

Languages: Java, C++, Python, JavaScript, SQL

Backend: Spring Boot, Spring Security, Spring Data JPA, REST API Design

Databases: MySQL, MongoDB, Redis, Vector Databases (ChromaDB)

DevOps & Tools: Docker, Docker Compose, Git, Maven, Postman, IntelliJ IDEA

CS Fundamentals: DSA, OOP, DBMS, Operating Systems, System Design Basics

Frontend: React.js

Projects

HireTrack – Job Application Tracking System | *Java, Spring Boot, MySQL, JWT* | [GitHub](#) 2026

- Architected a multi-role backend (USER, RECRUITER, ADMIN) using Spring Boot with JWT stateless authentication, BCrypt password hashing, and Spring Security for role-based endpoint authorization.
- Designed and implemented 10+ secure RESTful APIs managing application life cycles with DTO-based mapping to prevent entity over-exposure and optimize payload sizes.
- Applied global exception handling via `@ControllerAdvice` and strict Bean Validation across a layered Controller → Service → Repository architecture to handle edge-case payloads seamlessly.

URL Shortener – Full Stack | *Spring Boot, Redis, React, Docker* | [GitHub](#) 2025

- Built a high-concurrency URL shortener using Base62 encoding for collision-free short URL generation.
- Implemented a Redis cache-aside pattern for $O(1)$ redirect lookups, reducing database read latency by over 70% during simulated high-traffic spikes.
- Decoupled data-heavy metrics writes from the active redirect path via async batch writes using Spring Scheduler; containerized the entire application stack using Docker Compose.

Reverse Product Search – CLIP-based Visual Retrieval | *Python, CLIP ViT-B/32, ChromaDB, OpenAI* 2025

- Developed an image-based product discovery system using CLIP ViT-B/32 to generate high-dimensional embeddings from raw product datasets.
- Integrated a vector database layer using ChromaDB to store embeddings, implementing fast cosine similarity search to retrieve matching items with sub-100ms retrieval latency.

Experience & Leadership

Technical Lead – Sahyadri Robotics Club Sep 2025 – Present

Mentorship & Systems Engineering Design

Mangaluru, Karnataka

- Lead technical architecture and code reviews for 20+ students; direct hardware-software embedded communication logic and multi-sensor integration.
- Refined firmware loop execution times by benchmarking real-time sensor processing pipelines in C++, reducing system reaction lag.

Smart India Hackathon – ML for Cattle Breed Recognition Sep 2025

TensorFlow, Computer Vision Pipeline

- Built a TensorFlow-based image classification model to recognize cattle breeds from unstructured real-world data, orchestrating the full pipeline from image preprocessing to automated deployment.

Techkriti – IIT Kanpur National Robotics Competition Mar 2024

Embedded Motor Control & Firmware Optimization

- Designed and built Pick-and-Place and Line Following autonomous bots, optimizing PID motor control loops and IR sensor array logic in C++ to achieve precise stability.

Education

Sahyadri College of Engineering and Management

B.E. in Robotics and Automation Engineering; CGPA: 8.9/10

Mangaluru, Karnataka

2023 – Present (Expected 2027)

Certifications

Microsoft Certified – Azure Fundamentals (AZ-900), 2025 | Cisco Networking Academy – Networking Basics, 2026